Siqi Fan

Institute of Automation, CAS – Beijing, China

☎ +86 15201595986 ✉ leo_1@163.com 🗎 https://leofansq.github.io

Education

University of Chinese Academy of Sciences (UCAS) **Beijing, China**

- Master of Engineering in Automation *Sep. 2019 – Now*
- School of Artificial Intelligence

Shanghai Jiao Tong University (SJTU) **Shanghai, China**

- Bachelor of Engineering in Automation *Sep. 2015 - Jun. 2019*
- School of Electronic Information and Electrical Engineering

Publications

- **S. Fan**, Q. Dong, F. Zhu, et al. SCF-Net: Learning spatial contextual features for large-scale point cloud segmentation. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, (**CVPR**), 2021.
- **S. Fan**, F. Zhu, S. Chen, et al. FII-CenterNet: An anchor-free detector with foreground attention for traffic object detection. In IEEE Transactions on Vehicular Technology, (**IEEE TVT**), 2021.
- **S. Fan**, F. Zhu, H. Chang, et al. Improving road detection results based on ensemble learning and key samples focusing. In Proceedings of the IEEE International Conference on Intelligent Transportation Systems, (**ITSC**), 2020.

Experience

Institute of Automation, Chinese Academy of Sciences (CASIA) **Beijing, China**

Student Researcher, Computer Vision & Intelligent Vehicles *Sep. 2019 – Now*

- Works on the environment perception of the intelligent vehicles
  - **Introducing foreground attention to anchor-free detector for traffic object detection**
    - Proposed a foreground segmentation approach for anchor-free object detection, which could alleviate the background influences under the complex traffic environment with little extra computation cost.
    - Generated the box-induced labels of the foreground segmentation directly based on the bounding-box labels without additional labeling cost.
    - Achieved the advanced performance in both accuracy and efficiency.
  - **Novel learning framework for semi-supervised semantic segmentation**
    - Proposed the conservative-progressive collaborating learning framework, which achieves the collaboration of conservative evolution and progressive exploration.
    - Proposed a pseudo labeling approach based on the class-wise confusion indicator for the disagreement part of the paralleled branches.
    - Proposed a confidence-based dynamic loss to reduce the influences of the noisy label.

- Achieved the SOTA performance and the advantage is more evident when fewer labeled data is leveraged. The performance of full-supervision and few-supervision is improved.

- **Learning spatial contextual features for large scale point cloud segmentation**
  - Proposed a systematic approach for learning the spatial contextual feature, including local information representation, local and global spatial contextual feature learning.
  - Designed the corresponding module and a new 3D semantic segmentation network.
  - Achieved the SOTA performance for large scale point cloud semantic segmentation, and the performances on other related applications are verified by other researchers.
- **Engineering projects**: mapping and localization based on LiDAR, waypoints follow, lane follow based on camera, data interchange

Intel Labs China (ILC) **Beijing, China**

Autonomous System Research Intern, Intelligent Vehicles *Aug. 2020 – Dec. 2021*

- Worked on the safety research of the intelligent vehicles
  - **Simulation evaluation of the Responsibility -Sensitive Safety (RSS)**
    - Evaluation of RSS on CARLA: build the test system, generate the test scenes and scenarios, conduct the experiments, analyze the experimental results, write the evaluation reports
    - Evaluation of RSS on MATLAB: refine the test system, conduct the experiments, analyze the experimental results, write the evaluation reports
  - **Field test of the Responsibility -Sensitive Safety (RSS)**
    - Realized the test system: RTK-based perception, TCP-based transmission, CAN-based control
  - **Cross-team cooperation and academic communication with the partner universities**
  - **Participated in drafting the China ITS Industry Alliance Group Standard** "Technical Requirement of Safety Assurance of AV Decision Making".
  - The program was awarded **Intel's highest honor "Intel Achievement Awards (IAA)"**

Intelligent Vehicle Lab, Shanghai Jiao Tong University **Shanghai, China**

Student Researcher, Computer Vision & Intelligent Vehicles *Jul. 2017 – Jun. 2019*

- Worked on the object detection of the intelligent vehicles
  - **Obstacle detection system for LiDAR based on embedded device**
    - Initiated and led the "National Innovation Program for College Students".
  - **3D object detection based on the fusion of multi-modal information**

Honors and Awards

- **National Scholarship**, 2021
- **Shanghai Jiao Tong University Excellent Scholarship**, 2018
- **China Industrial Intelligence Challenge, Outstanding Award (State Level)**, 2018

Academic Services

- **Reviewer:** IEEE Transactions on Intelligent Vehicles, IET Computer Vision, IET Cyber-Systems and Robotics, IEEE Intelligent Transportation Systems Magazine